

Treatment Planning (antibiotics)

William just got an aggressive form of the seasonal flu, and the doctor prescribed him a broad-spectrum antibiotic to fight it, the *Panaceaxyl*TM. This antibiotic has to be assumed in a strictly controlled pattern, exactly once every T hours, until the flu is defeated: never stop a treatment early, or the illness will get much worse! Even the slightest transgression to the prescription, such as anticipating the timing by just a few minutes, might result in one of the numerous *Panaceaxyl*TM side effects.



Figure 1: List of *Panaceaxyl*TM side effects, as reported in 0.1 sec by the drug's commercial.

However, William is a very busy person, having already taken N commitments on his agenda, each on a different hour H_i . It would not be a good idea to plan the antibiotic for one of these hours: William is definitely going to forget about it if he is distracted by other business! More precisely, if William is going to start the treatment in hour X , he needs to have hours:

$$X, X + T, X + 2T, \dots, X + iT, \dots$$

free from commitments for all $i > 0$. Help William calculate the first hour in which he can start the treatment without risking the dreadful side effects.

Among the attachments of this task you may find a template file `antibiotics.*` with a sample incomplete implementation.

Input

The first line contains integers N, T . The second line contains N integers H_i .

Output

You need to write a single line with an integer: the smallest hour X that is safe for starting the treatment.

Constraints

- $1 \leq N \leq 100\,000$.
- $1 \leq T \leq 10^9$.
- $0 \leq H_i \leq 10^9$ for each $i = 0 \dots N - 1$.
- There are no repeated commitments.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** [5 points]: Examples.
- **Subtask 2** [10 points]: $T = 1$.
- **Subtask 3** [30 points]: $N, T, H_i \leq 1000$ for all i .
- **Subtask 4** [25 points]: $N, T \leq 1000$.
- **Subtask 5** [20 points]: $T \leq 1000$.
- **Subtask 6** [10 points]: No additional limitations.

Examples

input.txt	output.txt
6 8 16 1 4 10 15 12	3
9 3 16 9 20 11 13 3 0 12 2	15

Explanation

In the **first sample case**, starting hour 0 is not feasible due to the commitment in hour 16, hour 1 is already occupied and hour 2 is not feasible due to the commitment in hour 10. Starting hours 5 and 6 are also feasible but not optimal.

In the **second sample case**, any hour before 15 is not feasible due to one of the commitments in hours 12, 16 or 20.